

Reader Report: Week 05

Bayesian Invariant Prediction & Optimization-Based ICP

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Summary

Wu and Blei (2025) introduce **Bayesian Invariant Prediction** (BIP), casting ICP as latent-variable inference. A binary selector $z \in \{0, 1\}^p$ indexes invariant features; the posterior $p(z \mid \mathcal{D})$ is proportional to likelihood ratios between pooled and per-environment conditionals. They prove consistency (Thm 1) and exponential contraction at rate $O(R \cdot e^{-\kappa n E \mu_{\min}})$ (Thm 2), where $\mu_{\min}$ captures environment heterogeneity. For scalability, **VI-BIP** uses mean-field variational inference with the U2G gradient estimator; it identifies invariant genes in a 6,170-gene perturbation dataset where competing methods falter.

Fan et al. (2024) propose **NegDRO**, a continuous minimax formulation: $\min_b \max_{w \in U(\gamma)} \sum_e w_e \mathbb{E}[Y^e - b^\top X^e]^2$, where $U(\gamma)$ allows *negative weights* to enforce risk invariance. Under additive interventions, every stationary point is $O((1+\gamma|E|)^{-1} + T^{-1/4} + n^{-1/4})$ -close to β^* (Thm 5), enabling gradient descent in *polynomial* time rather than 2^p enumeration.

Contrast: BIP gives full posterior uncertainty over invariant sets; NegDRO trades distributional characterization for polynomial-time convergence with provable rates.

Critique & Project Connection

For my *Predicting NIMBYism Causally* project (~ 4 temporal environments, $p \approx 50$ features), BIP’s multi-modal posterior honestly quantifies which invariant sets remain ambiguous with limited environments. NegDRO’s polynomial-time guarantee and tolerance of hidden confounding (e.g., unmeasured wealth confounding school quality and opposition) better suit iterative model development. A hybrid strategy could use NegDRO for fast screening, then BIP for uncertainty quantification on a reduced feature set. **Limitation:** both assume linear/squared-loss models; extending to binary outcomes ($Y \in \{0, 1\}$) remains future work.

Questions

1. NegDRO identifies β^* when Condition 1b holds ($\lambda > 0$, second-moment heterogeneity); BIP’s posterior contracts when $\mu_{\min} > 0$ (KL heterogeneity between local and pooled conditionals). Can environments satisfy one measure but not the other—and if so, which method degrades more gracefully?
2. Both methods assume squared loss. For binary Y (as in protest data), the invariance guarantee rests on a misspecified loss. Does misspecification break symmetrically, or could NegDRO’s risk-parity constraint remain well-posed while BIP’s posterior concentrates on the wrong z^* ?

Project Update: Predicting NIMBYism Causally

1. What is the problem?

We seek the causal drivers of zoning opposition (NIMBYism) in Austin, TX. Following Peters et al. (2016), we adopt invariant causal prediction (ICP): find S^* such that $P_e(Y \mid X_{S^*}) = P(Y \mid X_{S^*})$ across all environments e . Features stable across regimes are identified as causal; those whose predictive power breaks are discarded as spurious.

The key to identification is environment diversity—each regime change stress-tests predictors, eliminating spurious ones. Over 2019–2024, Austin experienced the Adler-to-Watson mayoral transition (Jan 2023), parking-reform elimination (Nov 2023), the HOME Initiative (Feb 2024), and council-seat turnovers. These yield *temporal* environments (citywide reforms) and *geographic* ones (council-district turnovers; parcels outside Austin city limits as a natural control). Each event intervenes on a subset of covariates while preserving the causal mechanism (Bühlmann, 2020, assumption B), constituting **limited additive interventions** (Fan et al., 2024; Rothenhäusler et al., 2019). Peters et al.’s estimated set $\hat{S}(\mathcal{E}) \supseteq S^*$ is an outer bound on the true causal parents that shrinks monotonically with each additional environment (Peters et al., 2016, Thm 1); Fan et al.’s Condition 4 extends identification to partial interventions, and BIP’s posterior quantifies remaining ambiguity. With $p \approx 50$ features, classical ICP’s 2^p enumeration is infeasible, motivating both methods.

2. Mathematical Formulation

We formalize the environments from Section 1. Let $\mathcal{E} = \{e_1, \dots, e_K\}$ index the K regimes above. In each e we observe $(X_i^e, Y_i^e)_{i=1}^{n_e}$ with $X^e \in \mathbb{R}^p$ (valuations, lot dimensions, land-use class, tax exemptions) and $Y^e \in \{0, 1\}$ (protest signed). Covariates partition into intervened S_L (regulatory/market features) and untouched S_L^c (demographics, schools, transit).

This partition motivates the linear SEM (Peters et al., 2017, Ch. 4) under the additive intervention regime (Rothenhäusler et al., 2019):

$$Y^e = (\beta^*)^\top X_{S^*}^e + \varepsilon_Y^e, \quad \varepsilon_X^e \stackrel{d}{=} \eta_X + (0_{S_L^c}, \delta_{S_L}^e)^\top, \quad \mathbb{E}[\eta(\delta^e)^\top] = 0.$$

Here $S^* \subseteq [p]$ indexes the true causal parents; S_L the intervened covariates; S_L^c the untouched ones. Only $\delta_{S_L}^e$ varies, so $Y \mid X_{S^*}$ is stable and prediction risk $\mathbb{E}[(Y^e - (\beta^*)^\top X^e)^2] = \sigma_Y^2$ holds in every environment—the *risk invariance* we seek to exploit.

Two complementary methods recover S^*, β^* . Fan et al. (2024) cast the problem as a minimax optimization: $\hat{\beta}^\gamma \in \arg \min_b \max_{w \in U(\gamma)} \sum_e w_e \hat{\mathbb{E}}[(Y^e - b^\top X^e)^2]$, where $U(\gamma)$ allows *negative* weights to enforce risk parity; under Condition 4, this recovers β^* even with limited interventions. Wu and Blei (2025) take a Bayesian route: a latent selector $z \in \{0, 1\}^p$ with posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z) / \hat{p}_e(Y_i^e \mid X_{ei}^z)$ concentrates on z^* as data and environments grow, while quantifying remaining ambiguity.

References

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